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Asymptotics of Partition Functions of Coulomb Gases

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"En remontant chez moi pour y passer la soirée à travailler de mon mieux,
je me disais que le monde n'est pas construit pour l'équilibre.
Le monde est désordre.
L'équilibre n'est pas la règle, c'est l'exception.
Et je faisais le serment de travailler pour l'ordre et l'équilibre.
Que de serments! Tu vas sourire."
G.Duhamel, Maitres, 1937

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For all of those who are or were:

Your courage, patience, daring, and caring has brought me here.

To hoping that I become deserving of such act, for I feel loved and dear; of immense luck.

For all of those who will be:

To promise that I can for you as it was done for me.

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“João, é bom ver que você aprendeu e fez tantas coisas bonitas!
Continue assim, muito feliz! Brinque e aprenda muito.”
Te amo muito, Papai.

RESUMO

JOÃO V. A. PIMENTA. **Assintóticas para funções partição de gases de Coulomb**. 2026. 130 p. Dissertação (Mestrado em Ciências – Matemática) – Instituto de Ciências Matemáticas e de Computação, Universidade de São Paulo, São Carlos – SP, 2026.

A Teoria de Matrizes Aleatórias (RMT) tem sido uma teoria probabilística bem sucedida no estudo de sistemas complexos. Na década de 1950, Wigner propôs que as estatísticas locais dos níveis de ressonância de espalhamento de nêutrons em núcleos pesados poderiam ser modeladas pelo comportamento estatístico dos autovalores de grandes matrizes hermitianas com invariância ortogonal, unitária ou simplética. Desde então, físicos e matemáticos perceberam que poderiam usar as estatísticas de grandes matrizes para modelar as estatísticas de sistemas anteriormente inacessíveis, como sistemas muito grandes ou complexos. Embora na física a aplicação para o espalhamento de núcleos atômicos fosse direta, os estatísticos encontraram muitas aplicações para a teoria como no estudo de matrizes de correlação, e matemáticos fizeram grandes avanços ao conectar a RMT a sistemas integráveis. Focaremos nas distribuições de autovalores cujas funções de correlação podem ser expressas como o determinante de uma função matricial de polinomios ortogonais. A função partição é uma soma sobre os microestados deste sistema de partículas e, como tal, muitas das macroquantidades, como entropia e energia, são funções da mesma. Além disso, pode ser expressa em termos dos polinomios ortogonais e, por isso, frequentemente, as macroquantidades do sistema estão diretamente relacionadas com a assintótica dos mesmos polinomios ortogonais. Tal assintótica pode ser reduzida à técnica de Laplace se o polinomio ortogonal possuir a representação integral apropriada. Neste manuscrito, contextualizaremos e replicaremos uma importante e recente técnica para derivar a assintótica da chamada função partição de uma classe de conjuntos invariantes de matrizes aleatórias com autovalores complexos.

Palavras-chave: Teoria de Matrizes Aleatórias, Ensembles Invariante de Matrizes, Polinômios Ortogonais, Processos Pontuais, Assintótica.

ABSTRACT

JOÃO V. A. PIMENTA. **Asymptotics of Partition Functions of Coulomb Gases** . 2026. 130 p. Dissertação (Mestrado em Ciências – Matemática) – Instituto de Ciências Matemáticas e de Computação, Universidade de São Paulo, São Carlos – SP, 2026.

Random Matrix Theory (RMT) has been a successful probabilistic theory in the study of complex systems. In the 1950s, Wigner proposed that local the statistics of scattering resonance levels for neutrons off heavy nuclei could be modeled by the statistical behavior of eigenvalues of large hermitian matrices with orthogonal, unitary or symplectic invariance. Thenceforth physicist and mathematicians realized they could make use of the statistics of large matrices to model statistics of previously inaccessible systems, e.g. very large or complicated. Although in physics the application for atomic nuclei scattering was straightforward, statisticians found many uses of the theory in the study of correlation matrices and mathematicians made large advances linking RMT to integrable systems. We focus on eigenvalue distributions which correlation functions can be expressed as the determinant of a matricial function of orthogonal polynomials. The partition function is a summation over the micro-states of this particle system and as such, many of the macro-quantities, i.e. as entropy an energy, are a function of it. Moreover it can be expressed in terms of the orthogonal polynomials and therefore, oftentimes, macro-quantities of the system are directly related the asymptotic of orthogonal polynomials. Such asymptotics can be reduced to the Laplace technique if the orthogonal polynomial has a appropriate integral representation. In this manuscript we contextualize and replicate a important recent technique for deriving the asymptotics for the so-called partition function of a class of invariant ensembles of random matrices with complex eigenvalues.

Keywords: Random Matrix Theory, Invariant Matrix Ensembles, Orthogonal Polynomials, Point Processes, Asymptotics.

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INTRODUCTION

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RANDOM MATRIX THEORY

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ASYMPTOTICS

There is a sense in the previous chapters that we want to study distributions of eigenvalues when our matrices are large enough - be that for feasibility of computations or approximation of infinite dimension operators. We also have mentioned that our main interest here is the invariant ensembles, specially the orthogonal polynomial ensemble. Because of the expression of the univariate eigenvalue density as the determinant of a orthogonal polynomial kernel function, we change our focus to the study of these polynomials. Studying the distribution of eigenvalue for large matrices is equivalent, in this sense, to studying the regime of large n for the kernels and consequently, for the orthogonal polynomials.

For some of those polynomials, a integral representation is available, allowing us to use asymptotic methods on integrals to solve for the large n behavior. These methods revolve around the idea that we want to integrate a function that accumulates its value in some discrete set of points in the domain as we take some of its parameters to infinity. This is usually an exponential with negative exponent depending on some parameter n that we identify with the size of our initial matrices. Then, as n grows, the approximation of this integral as proportional to the value of the function on these point of maxima decreases in error. This method roughly describes the *Steepest Descent* technique, which will be described in this chapter. The sections that precedes this descriptions are the building blocks for the more sophisticated technique; we go through *Watson's lemma* and the *Laplace technique*.

Before going through asymptotic techniques in integrals, we briefly describe a technical result we will make use of in Chapter 5. This is a quadrature technique or, more specifically, the *Euler-Maclaurin* formula. This formula will later allow us to approximate very complicated summations as integrals that we can compute analytically. Techniques and structuring here are based on the book by Olver *et al.* (2010) and Alpert (1999) for quadrature and on the lecture notes by Silva (2021) for asymptotics on integrals.

4.1 Quadrature Technique

Before doing asymptotics on integrals we discuss a technique that will become important when approximating large summations: quadratures, i.e. approximating integrals by summations and vice-versa. One can see that asymptotics on integrals can then also perform a role on discrete summations. We first start briefly introducing a useful tool in the development of the formula we wish to use: the Bernoulli numbers B_n, polynomials $B_n(x)$ and periodic polynomials $\tilde{B}_n(x)$. The Bernoulli numbers B_n are a sequence of rational numbers that frequently appears in the Taylor series expansion of trigonometric-type functions. These number can also be seen as special values obtained by the Bernoulli polynomials $B_n(x)$ when evaluated at 0, i.e. $B_n(0) = B_n$. We also can express the expansion formula for these polynomials given by

$$B_n(x+h) = \sum_{k=0}^{n} \binom{n}{k} B_k(x) h^{n-k},$$

that give us that $B_0(x) = 1$. More interestingly for us is the fact that these polynomials satisfy the differential equation

$$B_n'(x) = nB_{n-1}(x), \quad n = 1,2,\cdots \tag{4.1}$$

Finally, the periodic Bernoulli polynomial is simply the Bernoulli polynomials evaluated at the fractional part of the argument, i.e., $\tilde{B}_n(x) = B_n(x - \lfloor x \rfloor)$. Now, let B_n and $B_n(x)$ denote the n-th Bernoulli number and polynomial, respectively, and $\tilde{B}_n(x)$ the n-th Bernoulli periodic function $B_n(x - \lfloor x \rfloor)$. Assume that a, b are reals and p is an integer such that $b > a$, $p > 0$, and $f^{(2k)}(x)$ is absolutely integrable over $[a,b]$. Then,

Theorem 4.1. *(Euler-Maclaurin Formula, (ALPERT, 1999)) For some function $f \in C^p(\mathbb{R})$ and $p \geq 1$ let $h = (b-a)/n$. Then, the Euler-Maclaurin formula for the interval $[a,b]$ is given by*

$$h\sum_{j=1}^{n-1} f(a+jh) = \int_a^b f(x)\mathrm{d}x - \frac{h}{2}(f(b)-f(a)) + \sum_{r=1}^{p-1} \frac{h^{r+1}B_{r+1}}{(r+1)!}\left[f^{(r)}(b) - f^{(r)}(a)\right]$$
$$- h^{p+1}\int_0^1 \frac{B_p(1-x)}{(p)!} f^{(p)}(c+xh)\mathrm{d}x$$

This identity is called the *Euler-Maclaurin* formula. It can be extremely useful when estimating the asymptotic of the summation on the function f when its integral can be more easily evaluate. In essence, this is a quadrature technique; a numerical integration: the approximation of an integral $\int f(x)\mathrm{d}x$ of some function f by a discrete summation $\sum w_i f(x_i)$ over a set of points x_i with some weight function w_i. For example, in the classical Riemann sums, one approximates the integral $\int f(x)\mathrm{d}x$ over some interval $[a,b]$ with the limit of the summation $\sum f(x_i^*)\Delta x_i$, where $x_i^* \in [x_{i-1}, x_i]$ and $\Delta x_i = x_i - x_{i-1}$. Different methods can be generated by choosing x_i^* differently.

The trapezoidal rule, for example, is a Riemann sum where x_i^* is chosen to be the mean point x_i^* is such that $f(x_i^*) = (f(x_{i-1}) + f(x_i))/2$. In this way, we would estimate the area using trapezoids, as the name suggests. Then, if $h = (b-a)/2$, we write

$$\begin{aligned}\int_a^b f(x)\mathrm{d}x &\approx h\sum_{j=m}^{n-1} f(j^*) = h\sum_{j=m}^{n-1}\frac{f(j+1)+f(j)}{2},\\ &= h\sum_{j=m+1}^{n-1} f(j) + \frac{h}{2}(f(n)+f(m)) = h\sum_{j=m}^{n-1} f(j) + \frac{h}{2}(f(n)-f(m)).\end{aligned}$$

Of course, this has some error term. Suppose that for some (x_i, x_{i-1}) we have the same k-derivatives at both extreme points for every $k \geq 1$, i.e., the function is constant. Then, there is no error in the approximation. If however, the two extreme points differ only on the k-th derivative, there is some curvature defined by this difference where the error comes from. Therefore, there is indication of some error function given in respect to higher order derivatives.

Proof. Following the computations in Alpert (1999), take a function $f \in C^p(\mathbb{R})$ for $p \geq 1$. To derive the Euler-Maclaurin for the interval $[a,b]$, let $h = (b-a)/n$ and $c = a + jh$, then we express

$$\int_a^b f(x)\mathrm{d}x = \sum_{j=0}^{n-1}\int_{a+jh}^{a+(j+1)h} f(x)\mathrm{d}x$$

where we use (4.1) and integration by parts to rewrite each integral as

$$\begin{aligned}\int_c^{c+h} f(x)\mathrm{d}x &= h\int_0^1 B_0(1-x)f(c+xh)\mathrm{d}x\\ &= -h\frac{B_1(1-x)}{1!}f(c+xh)|_0^1 + h^2\int_0^1 \frac{B_1(1-x)}{1!}f'(c+xh)\mathrm{d}x,\\ &\vdots\\ &= -\sum_{r=0}^{p-1}\frac{h^{r+1}B_{r+1}(1-x)}{(r+1)!}f^{(r)}(c+xh)|_0^1 + h^{p+1}\int_0^1\frac{h^{r+1}B_p(1-x)}{(p)!}f^{(p)}(c+xh)\mathrm{d}x,\\ &= h\frac{f(c)+f(c+h)}{2} - \sum_{r=1}^{p-1}\frac{h^{r+1}B_{r+1}}{(r+1)!}\left[f^{(r)}(c+h) - f^{(r)}(c)\right]\\ &\qquad + h^{p+1}\int_0^1\frac{B_p(1-x)}{(p)!}f^{(p)}(c+xh)\mathrm{d}x.\end{aligned}$$

where one can start to notice, for instance, the trapezoidal rule creeping out on the first term. Notice that it was necessary to use that $-B_1(0) = B_1(1) = 1/2$ and $B_n(0) = B_n(1) = B_n$ for $n \neq 1$. Returning to the summation, we rearrange terms as

$$\begin{aligned}h\left[\frac{f(a)}{2} + f(a+h) + \cdots + f(b-h) + \frac{f(b)}{2}\right] &= \frac{h}{2}(f(b)-f(a)) + h\sum_{j=1}^{n-1} f(a+jh),\\ = \int_a^b f(x)\mathrm{d}x + \sum_{r=1}^{p-1}\frac{h^{r+1}B_{r+1}}{(r+1)!}\left[f^{(r)}(b) - f^{(r)}(a)\right] &- h^{p+1}\int_0^1\frac{B_p(1-x)}{(p)!}f^{(p)}(c+xh)\mathrm{d}x.\end{aligned}$$

□

4.2 Asymptotic on Integrals

The techniques we will expand on here refers to a specific type of complex function $\phi : \mathbb{C} \longrightarrow \mathbb{R}$ in the regime where $\lambda \longrightarrow \infty$. Those functions are of the form

$$\phi(\lambda) := \int_\Gamma g(s)\, \mathrm{e}^{-\lambda f(s)}\, \mathrm{d}s, \tag{4.2}$$

where Γ is a given contour in the plane and f, g are suitable functions. Suitable here means that we impose g and f to be continuous (or analytic); g is integrable; f have global minima with zero derivative and strictly positive second derivative at this point of minima. There is also a more technical regularity condition on f to ensure the approximations hold to still be made clear.

Example 4.1. (Airy Equation) Function such as in (4.2) are not uncommon. Consider the function that satisfies the equation $\psi''(x) = x\psi(x)$. Taking the Fourrier transform we have

$$-\omega^2 \Phi(x) = \mathrm{i}\Phi'(x)$$

for which we solve with $\Phi(x) = \mathrm{e}^{-\mathrm{i}\omega^3/3}$ and then take the inverse Fourrier Transform to get

$$\phi(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \mathrm{e}^{\mathrm{i}/3\omega^3 + \mathrm{i}x\omega}\, \mathrm{d}\omega, \quad x \in \mathbb{R}.$$

The imaginary part of the argument is an odd function and therefore zero when integrated. The real part is expressed

$$\phi(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \cos\left(\frac{\omega^3}{3} + x\omega\right) \mathrm{d}\omega.$$

This is the Airy function. For this integral, the techniques applied in this chapter are of great use to understand its behavior as $\omega \longrightarrow \infty$.

Consider $\phi : \mathbb{R} \longrightarrow \mathbb{R}$, f with a single global minima and $\Gamma \equiv \mathbb{R}$ - for simplicity; the argument's idea should hold for other choices given the appropriate considerations. If ϕ is as in (4.2), and satisfying the conditions given for such function, then take x^* the single minimal critical point for f. We first note that if that is true, then, for any $|\epsilon| > 0$, the following limit should hold:

$$\lim_{\lambda \to \infty} \frac{\mathrm{e}^{-\lambda(f(x^*))}}{\mathrm{e}^{-\lambda(f(x^* - \epsilon))}} = \lim_{\lambda \to \infty} \mathrm{e}^{-\lambda(f(x^*) - f(x^* - \epsilon))} = \lim_{\lambda \to \infty} \mathrm{e}^{\lambda |c|} \longrightarrow +\infty.$$

That is, the main contribution for the integral comes from a single point. The main idea then is to separate the integral in two regions: A first interval $(x^* - c/\sqrt{\lambda}, x^* + c/\sqrt{\lambda}), c > 0$, including this critical point and another interval that in theory should contribute only in error for the integral asymptotic. We start by approximating f around its global maximum by Taylor's theorem

$$f(x) = f(x^*) - \frac{1}{2}|f''(x^*)|(x-x^*)^2 + \epsilon(x-x^*)$$

and then perform a change of variables $\frac{x}{\sqrt{\lambda}} = s - x^*$, so that

$$\int_{x^*-\frac{c}{\sqrt{\lambda}}}^{x^*+\frac{c}{\sqrt{\lambda}}} g(s)\,\mathrm{e}^{-\lambda f(s)}\,\mathrm{d}(s) = \frac{\mathrm{e}^{-\lambda f(x^*)}}{\sqrt{\lambda}} \int_{-c}^{c} g\left(x^* + \frac{x}{\sqrt{\lambda}}\right) \mathrm{e}^{\frac{1}{2}|f''(x^*)|x^2 - \epsilon\left(\frac{x^3}{\sqrt{\lambda}}\right)}\,\mathrm{d}x.$$

Now, using the technical assumption of regularity on the function f, one can argue that the error term ϵ goes to zero sufficiently fast, leaving a Gaussian Integral. Moreover, one needs to argue that the integral outside this small neighborhood of x^* is exponentially smaller then $\mathrm{e}^{-\lambda f(x^*)}$. This should follow by an argument where the asymptotic of the integral

$$\int_{x^*+\frac{c}{\sqrt{\lambda}}}^{\infty} |g(s)|\,\mathrm{e}^{-\lambda f(s)}\,\mathrm{d}(s) \le \mathrm{e}^{-\lambda f(x^*+c/\sqrt{\lambda})}\,||g||_{L^1}$$

is exponentially smaller than the previous region.

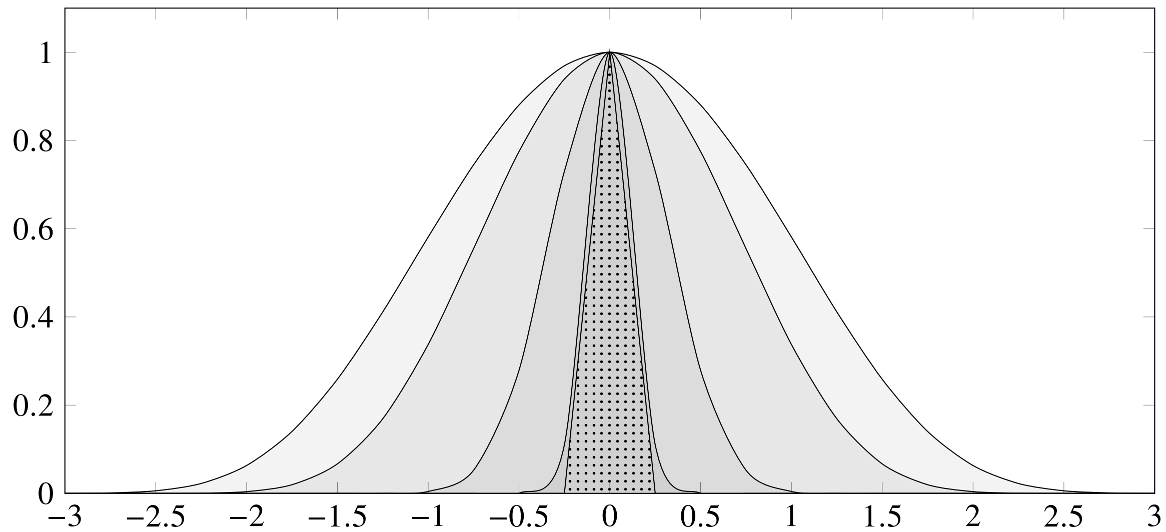

Figure 1 – In the Plot it is represented the plot for $\mathrm{e}^{-\lambda(\cosh(x)-1)}$ and an increasing $\lambda = 1, 2, 10, 70$ - darker represent larger values of the parameter. The dotted patter represents $\mathrm{e}^{-\lambda x^2}$ for large lambda.

With this, we only note a few details of interest. Firstly, more precise error estimates can be obtained if one uses higher order terms of the Taylor Series expansion of f. For maximizer x^* which have null second derivative (a double critical point), the Laplace Approximation can be used expanding to the third order terms of the Taylor Series $f(x) = f(x^*) = \frac{1}{6}f'''(x^*)(x-x^*)^3 + \epsilon(x-x^*)$ and making the change of variables $t/\sqrt[3]{n} = x - x^*$ instead. This argument and result could also be used on curves on the plane. This is motivated from the fact that every (smooth) curve is locally a straight line. This will lead us to the method of *Steepest Descent*. These next subsections are in the spirit of proving a bit more formally the expansion given by the *Laplace Method* and introducing the mentioned *Steepest Descent*.

4.2.1 Watson's Lemma

We start this topic on the real line by looking at the functions $F : \mathbb{R} \to \mathbb{R}$ of the form

$$F(\lambda) = \int_0^T f(t)\, \mathrm{e}^{-\lambda t}\, \mathrm{d}t, \tag{4.3}$$

which one may think of as the Laplace transform of a function f supported on $[0,T]$. Note that F fits well with the models we described previously, and so one should expect the main contribution to come from the minima of the exponent function. Let us formalize the conditions for such expectation and the consequent result.

Theorem 4.2. *(Watson's Lemma) Take $f \in L^1(0,T)$ and suppose that for some $\delta > 0$ it is of the form*

$$f(t) = t^{\alpha} f_0(t), \quad 0 < t \leq \delta,$$

where $\alpha \in (-1,\infty)$ and f_0 is continuously differentiable on $[0,\delta]$ with $f_0(0) \neq 0$. Then the function F in (4.3) satisfies

$$F(\lambda) = \frac{f_0(0)\Gamma(\alpha+1)}{\lambda^{\alpha+1}}(1 + O(\lambda^{-1}))$$

when $\lambda \to \infty$. Also, Γ is the usual Gamma function.

We must first mention that a weak estimation of the integral is given by

$$\int_0^T f(t)\, \mathrm{e}^{-\lambda t}\, \mathrm{d}t \leq ||f||_{L^1(0,T)}$$

where we only use the fact that $f \in L^1(0,T)$. However, we have more information on the behavior of the function in the section $0 < t < \delta$. Because of that fact, it is expected that in this set we could refine the approximation.

Proof. We start by decomposing the function in two parts

$$F(\lambda) = \int_0^{\delta} f(t)\, \mathrm{e}^{-\lambda t}\, \mathrm{d}t + \int_{\delta}^T f(t)\, \mathrm{e}^{-\lambda t}\, \mathrm{d}t =: (1) + (2).$$

For (2), we use the same weak estimate as commented before,

$$\int_{\delta}^T f(t)\, \mathrm{e}^{-\lambda t}\, \mathrm{d}t \leq \mathrm{e}^{-\lambda\delta} \int_{\delta}^T |f(t)| \mathrm{d}t = O(\mathrm{e}^{-\lambda\delta}\, ||f||_{L^1(\delta,T)}).$$

We keep it this way as we have no further information on the f in this section of the real line. This leave us with

$$F(\lambda) = \int_0^{\delta} f(t)\, \mathrm{e}^{-\lambda t}\, \mathrm{d}t + O(\mathrm{e}^{-\lambda\delta}\, ||f||_{L^1(\delta,T)}).$$

Now, we remind of the condition of continuous differentiability and the hypothesis on $f(t)$ to get $f(t) = t^{\alpha}(f_0(0) + r(t))$ for some differentiable error function $r(t)$ satisfying $|r(t)| \leq ||f_0'||_{L^\infty(0,\delta)} t$

with $||f_0'||_{L^\infty(0,\delta)} = \inf\{c \geq 0 \mid \mu|_{(0,\delta)}(\{|f_0'| > c\}) = 0\}$ as usual. Then, write (1) as

$$(1) = \int_0^\delta t^\alpha (f_0(0) + r(t)) \,\mathrm{e}^{-\lambda t} \,\mathrm{d}t = f_0(0) \int_0^\delta t^\alpha \,\mathrm{e}^{-\lambda t} \,\mathrm{d}t + \int_0^\delta t^\alpha r(t) \,\mathrm{e}^{-\lambda t} \,\mathrm{d}t.$$

For these two integrals we simplify the problem by rewriting

$$\begin{aligned}
\int_0^\delta t^\alpha \,\mathrm{e}^{-\lambda t} &= \int_0^\infty t^\alpha \,\mathrm{e}^{-\lambda t} \,\mathrm{d}t - \int_\delta^\infty t^\alpha \,\mathrm{e}^{-\lambda t} \,\mathrm{d}t = \frac{\Gamma(\alpha+1)}{\lambda^{\alpha+1}} - \int_\delta^\infty t^\alpha \,\mathrm{e}^{-\lambda t} \,\mathrm{d}t, \\
\left| \int_0^\delta t^\alpha r(t) \,\mathrm{e}^{-\lambda t} \,\mathrm{d}t \right| &\leq ||f_0'||_{L^\infty(0,\delta)} \int_0^\delta t^{\alpha+1} \,\mathrm{e}^{-\lambda t} \,\mathrm{d}t, \\
&= ||f_0'||_{L^\infty(0,\delta)} \left(\int_0^\infty t^{\alpha+1} \,\mathrm{e}^{-\lambda t} \,\mathrm{d}t - \int_\delta^\infty t^{\alpha+1} \,\mathrm{e}^{-\lambda t} \,\mathrm{d}t \right), \\
&= ||f_0'||_{L^\infty(0,\delta)} \left(\frac{\Gamma(\alpha+2)}{\lambda^{\alpha+2}} - \int_\delta^\infty t^{\alpha+1} \,\mathrm{e}^{-\lambda t} \,\mathrm{d}t \right).
\end{aligned}$$

We are now only missing one ingredient to get the result, the order of

$$G(\lambda, \beta) := \int_\delta^\infty t^\beta \,\mathrm{e}^{-\lambda t} \,\mathrm{d}t, \quad \lambda > 1, \quad \beta > -1.$$

Using Cauchy-Schwartz we estimate

$$\begin{aligned}
|G(\lambda, \beta)|^2 &\leq \left(\int_\delta^\infty \mathrm{e}^{-\lambda t} \,\mathrm{d}t \right) \left(\int_\delta^\infty t^{2\beta} \,\mathrm{e}^{-\lambda t} \,\mathrm{d}t \right), \\
&= \left(\frac{e^{-\lambda\delta}}{\lambda} \right) \left(\int_\delta^\infty t^{2\beta} \,\mathrm{e}^{-\lambda t} \,\mathrm{d}t \right), \\
&\overset{(u=\lambda t)}{=} \left(\frac{e^{-\lambda\delta}}{\lambda} \right) \left(\int_\delta^\infty \mathrm{e}^{-u} u^{2\beta} \frac{1}{\lambda^{2\beta+1}} \mathrm{d}u \right), \\
&= \frac{\mathrm{e}^{-\lambda\delta}}{\lambda^{2\beta+2}} M_\beta
\end{aligned}$$

where M_β is some constant independent of λ. That leave us with

$$G(\lambda, \beta) = O\left(\frac{\mathrm{e}^{-\lambda\delta/2}}{\lambda^{\beta+1}} \right) = O\left(\mathrm{e}^{-\epsilon\lambda} \right)$$

where $\epsilon > 0$ depends on β and δ. Finally we rewrite

$$\int_0^\delta t^\alpha \,\mathrm{e}^{-\lambda t} = \frac{\Gamma(\alpha+1)}{\lambda^{\alpha+1}} - G(\lambda, \alpha) = \frac{\Gamma(\alpha+1)}{\lambda^{\alpha+1}} + O\left(\mathrm{e}^{-\epsilon\lambda} \right),$$

$$\begin{aligned}
\left| \int_0^\delta t^\alpha r(t) \,\mathrm{e}^{-\lambda t} \,\mathrm{d}t \right| &\leq ||f'||_{L^\infty(0,\delta)} \left(\frac{\Gamma(\alpha+2)}{\lambda^{\alpha+2}} - G(\lambda, \alpha+1) \right) \\
&= ||f'||_{L^\infty(0,\delta)} \left(\frac{\Gamma(\alpha+2)}{\lambda^{\alpha+2}} + O\left(\mathrm{e}^{-\epsilon\lambda} \right) \right) = O\left(\lambda^{-\alpha-2} \right),
\end{aligned}$$

and

$$F(\lambda) = O\left(\lambda^{-\alpha-2} \right) + f_0(0) \left(\frac{\Gamma(\alpha+1)}{\lambda^{\alpha+1}} + O\left(\mathrm{e}^{-\epsilon\lambda} \right) \right) + ||f||_{L^1(\delta,T)} \,\mathrm{e}^{-\delta\lambda}.$$

By the dominance of the exponential error, we get the result

$$F(\lambda) = \frac{f_0(0)\Gamma(\alpha+1)}{\lambda^{\alpha+1}}(1+O(\lambda^{-1})).$$

□

In the estimation above, the Gamma function and the factor of λ^{-1} is the result of integrating the linear exponent of $\mathrm{e}^{-\lambda t}$. If we, however, change this exponent to be of the general type $-\lambda t^d$ we still have a similar result on the behavior of the function.

Proposition 4.1. *For $T, S > 0$, define the function*

$$G(\lambda) := \int_{-S}^{T} g(t)\,\mathrm{e}^{-\lambda t^2}\,\mathrm{d}t$$

and assume $g(x)$ absolutely integrable in $[-S, T]$, of class C^1 in a fixed neighborhood of the origin. Then the function satisfies

$$G(\lambda) = \frac{g(0)\Gamma(1/2)}{\lambda^{1/2}} + O\left(\frac{1}{\lambda^{3/2}}\right)$$

when $\lambda \to \infty$ and where Γ is the Gamma function.

Proof. The proof is a reduction of the problem to Watson's Lemma. Assume, $T > S$ and write

$$G(\lambda) = \int_{-S}^{S} g(t)\,\mathrm{e}^{-\lambda t^2}\,\mathrm{d}t + \int_{S}^{T} g(t)\,\mathrm{e}^{-\lambda t^2}\,\mathrm{d}t.$$

The second integral we know to be of order

$$\int_{S}^{T} |g(t)|\,\mathrm{e}^{-\lambda t^2}\,\mathrm{d}t \le ||g||_\infty\,\mathrm{e}^{-\lambda S^2},$$

exponential on λ, so we drop it. For the first integral we rewrite

$$\int_{-S}^{S} g(t)\,\mathrm{e}^{-\lambda t^2}\,\mathrm{d}t = \int_{0}^{S} (g(t)+g(-t))\,\mathrm{e}^{-\lambda t^2}\,\mathrm{d}t,$$
$$\overset{(u=t^2)}{=} \frac{1}{2}\int_{0}^{S^2} \left(\frac{g\left(\sqrt{u}\right)+g\left(-\sqrt{u}\right)}{\sqrt{u}}\right)\mathrm{e}^{-\lambda u}\,\mathrm{d}u$$

where now we use Watson's Lemma to finish the proof. □

Example 4.2. Take the function called Gamma given by the expression

$$\Gamma(x) = \int_{0}^{\infty} \mathrm{e}^{-t}\,t^{x-1}\mathrm{d}t. \tag{4.4}$$

We are interested in what happens to the function when we take $x \to +\infty$. To see what happens, let us fit it into Equation 4.2. Rewrite (4.4) as

$$\int_0^\infty e^{-t+x\log(t)-\log(t)}\,dt.$$

This lead us to identify ϕ as $-\log(t)$, which would give us $\phi'(t)<0$, that is, we would have its minimum at $t=+\infty$. However $e^{\log(t)}=t^1$ is not integrable near the origin. Something need to change about the identification. Let us introduce a variable to rescale the integral: $xs=t$, s.t.,

$$\Gamma(x)=x^x\int_0^\infty e^{-\lambda(s-\log(s))}\,e^{-s}\,ds,$$

where we have used $\lambda=x-1$. Thus we identify $\phi(s)=s-\log(s)$ and $g(s)=e^{-s}$. In fact, now ϕ has a global minimum at $s=1$ and $\phi''(1)=1\neq 0$. Hence

$$\begin{aligned}\Gamma(x)&=\frac{\sqrt{2\pi}x^x\,e^{-x}}{(x-1)^{(-1/2)}}(1+O(x^{-1})),\\&=\sqrt{2\pi}x^{x+1/2}\,e^{-x}(1+O(x^{-1})),\end{aligned}$$

which is called the Stirling formula.

4.2.2 Laplace Method

A seemly extension of the type of function we were dealing with are functions of the type

$$\Phi(\lambda):=\int_a^b g(t)\,e^{-\lambda\phi(t)}\,dt \tag{4.5}$$

for $a,b\in\mathbb{R}\cup\{-\infty,\infty\}$ and $a<b$ and a well behaved function g. For now, ϕ is a real-valued function. To begin the study let us make some assumptions. The first one is that a is finite, that means that we set $a=0$ without loss of generality. The second assumption is that ϕ is a increasing function on $[0,b]$, so that $\phi(0)<\phi(t)$ for any t in the interval. With that we note that the coefficient $e^{-\lambda\phi(t)}/e^{-\lambda\phi(0)}$ decreases exponentially, that is, the integral is accumulated as the neighborhood of 0, i.e.

$$\Phi(\lambda)\approx\int_0^\delta g(t)\,e^{-\lambda\phi(t)}\,dt.$$

As we know $\phi(t)$ to be increasing, its first non zero derivative must be of even order, so that when we expanded the function by Taylor,

$$\phi(t)\approx\phi(0)+\phi_{(2k+1)}(0)t^{2k+1},$$

for some $k \geq 0$ and $|t| < \delta$ where $\phi_{(2k+1)} := \phi_{(2k+1)}/(2k+1)!$. As ϕ_{2k+1}, the $(2k+1)$th-derivative, must be positive we also have that, as $\lambda \to \infty$,

$$\begin{aligned}
\Phi(\lambda) &\approx e^{-\lambda\phi(0)} \int_0^\delta g(t)\, e^{-\lambda\phi_{(2k+1)}(t)\cdot t^{2k+1}}\, dt, \\
&\overset{(u=t(\lambda\phi_{(2k+1)}(0))^{1/(2k+1)})}{\approx} e^{-\lambda\phi(0)} \frac{g(0)}{(\lambda\phi_{(2k+1)}(0))^{\frac{1}{2k+1}}} \int_0^\infty e^{-u^{2k+1}}\, du, \\
&\overset{(s=u^{2k+1})}{\approx} e^{-\lambda\phi(0)} \frac{g(0)}{(2k+1)(\lambda\phi_{(2k+1)}(0))^{\frac{1}{2k+1}}} \int_0^\infty e^{-s}\, ds.
\end{aligned}$$

where the integral can be evaluated as a Gamma function. More than anything we need to focus on an important assumption that was made: that ϕ has a global minima at one of the endpoints of integration. If that were not the case, for example if $a < 0 < b$ and $\phi(t)$ had a global minima at $t = 0$ we now have to consider the expansion

$$\phi(t) \approx \phi(0) + \phi_{2k}(0)t^{2k}$$

which lead us to the expression

$$\Phi(\lambda) \approx e^{-\lambda\phi(0)} \frac{g(0)}{(\lambda\phi_{2k}(0))^{\frac{1}{2k}}} \int_{-\infty}^\infty e^{-u^{2k}}\, du.$$

This differentiates of the previously result more importantly by the limits of integration. This is closely related to physical boundaries of the problem. Before enunciating the proper Laplace Method we state the Implicit Function Theorem, which will become important in a series of results that follows.

Theorem 4.3. *(Implicit Function Theorem) Assume that S is an open subset of $\mathbb{R}^2$ and that $F : S \to \mathbb{R}$ is a function of class C^1. Assume also that (a, b) is a point in S such that*

$$F(a,b) = 0, \quad \text{and} \quad D_y F(a,b) \neq 0.$$

Then

1. *There exist $r_0, r_1 > 0$ such hat for every $x \in \mathbb{R}$ such that $|x - a| < r_0$, there exists a unique $y \in \mathbb{R}$ such that $|y - b| < r_1$*

   $$F(x,y) = 0.$$

   That is, defines a function $y = f(x)$ for $x \in \mathbb{R}$ near a and y close to b.

2. *Moreover, the function $f : B(r_0, a) \to B(r_1, b) \subset \mathbb{R}$ from item 1 is of class C^1, and its derivatives may be determined by differentiating the identity*

   $$F(x, f(x)) = 0$$

   and solving to find the partial derivatives of f.

Now we turn to our main interest.

Theorem 4.4. *(Laplace's Method, endpoint contributions) Let Φ be as in (4.5), with a finite and $g \in L^1(a,b)$. Assume that ϕ has a unique global minimum at $t = a$. In addition, suppose that for some $\delta > 0$ and some $\alpha > -1$, the function g takes the form*

$$g(t) = (t-a)^\alpha g_0(t)$$

for $|t-a| < \delta$ and g_0 continuously differentiable in $[a, a+\delta]$ and $g_0(a) \neq 0$, whereas ϕ is twice continuously differentiable on $[a, a+\delta]$ with $\phi'(a) > 0$. Then the function Φ satisfies an estimate of the form

$$\Phi(\lambda) = \frac{1}{\lambda^{\alpha+1}} \frac{g_0(a)\Gamma(\alpha+1)}{\phi'(a)^{\alpha+1}} \mathrm{e}^{-\lambda\phi(a)} \left(1 + O(\lambda^{-1})\right)$$

as $\lambda \to \infty$.

Proof. Start by assuming, without loss of generality, that $a = 0$ and dividing the integral in two intervals, one in the region where we have information about the behavior of g and another in the rest of the interval,

$$\Phi(\lambda) = \int_0^b g(t)\,\mathrm{e}^{-\lambda\phi(t)}\,\mathrm{d}t = \int_0^\delta g(t)\,\mathrm{e}^{-\lambda\phi(t)}\,\mathrm{d}t + \int_\delta^b g(t)\,\mathrm{e}^{-\lambda\phi(t)}\,\mathrm{d}t = (1) + (2).$$

Knowing that ϕ has a global minima at $t = 0$ we assume, changing δ as needed, that $\phi(t) > \phi(\delta)$ for any $t > \delta$. That give us a simple bound for (2):

$$|(2)| \leq \int_\delta^b |g(t)|\,\mathrm{e}^{-\lambda\phi(t)}\,\mathrm{d}t \leq \mathrm{e}^{-\lambda\phi(\delta)} \int_\delta^b |g(t)|\mathrm{d}t \leq \mathrm{e}^{-\lambda\phi(\delta)}\,||g||_{L^1[0,b]}.$$

We now get back to (1) and start by studying the difference given by $\phi(t) - \phi(0) = s$ in a similar way that we would think of a change of variable from t to s. Define the multivariate function

$$H(t,s) := \phi(t) - \phi(0) - s$$

such that $H(0,0) = 0$ and $\partial_t H(0,0) \neq 0$. This means we are in the condition to use Theorem 4.3 to say that there exists a unique bijective and continuously differentiable function $t(s) := t$ such that

$$\begin{cases} \phi(t(s)) - \phi(0) - s = 0, \\ t'(0) = \frac{\partial_s H(0,0)}{\partial_t H(0,0)} = \frac{1}{\phi'(0)}. \end{cases}$$

If we set $T = t^{-1}(\delta)$ we then have for (1)

$$\int_0^\delta g(t)\,\mathrm{e}^{-\lambda\phi(t)}\,\mathrm{d}t = \int_0^T g(t(s))\,\mathrm{e}^{-\lambda\phi(t(s))}\,t'(s)\mathrm{d}s.$$

Now, to apply the assumption on g we write $g(t(s)) = t^\alpha(s) g_0(t(s))$. As t is C^2 and vanishes at the origin, using its Taylor expansion, we have

$$t(s) = st'(0)(1 + r(s)) = \frac{s}{\phi'(0)}(1 + r(s)).$$

Joining both results,

$$\begin{aligned}
\int_0^T g(t(s))\,\mathrm{e}^{-\lambda\phi(t(s))}\,t'(s)\mathrm{d}s &= \mathrm{e}^{\lambda\phi(0)}\int_0^T g(t(s))\,\mathrm{e}^{-\lambda(\phi(t(s))-\phi(0))}\,t'(s)\mathrm{d}s \\
&= \mathrm{e}^{\lambda\phi(0)}\int_0^T t^{\alpha}(s)g_0(t(s))\,\mathrm{e}^{-\lambda s}\,t'(s)\mathrm{d}s \\
&= \frac{\mathrm{e}^{\lambda\phi(0)}}{\phi'(0)^{\alpha}}\int_0^T s^{\alpha}(1+r(s))^{\alpha}g_0(t(s))\,\mathrm{e}^{-\lambda s}\,t'(s)\mathrm{d}s
\end{aligned}$$

where, if we set $f(s) := s^{\alpha}(1+r(s))^{\alpha}g_0(t(s))$ we apply Watson's Lemma to finish the result. □

Theorem 4.5. *(Laplace's Method, interior contributions) Let Φ be as in (4.5), with a finite and $g \in L^1(a,b)$. Assume that ϕ has a unique global minimum at $t = c \in (a,b)$. In addition, suppose that for some $\delta > 0$ and some $\alpha > -1$, the function g takes the form*

$$g(t) = (t-a)^{\alpha}g_0(t)$$

for $|t-a| < \delta$ and g_0 continuously differentiable in $[c-\delta, c+\delta]$ and $g_0(a) \neq 0$, whereas ϕ is three times continuously differentiable on $[c-\delta, c+\delta]$ with $\phi'(a) > 0$. Then the function Φ satisfies an estimate of the form

$$\Phi(\lambda) = \mathrm{e}^{-\lambda\phi(c)}\left(\frac{\sqrt{2\pi}g(c)}{\sqrt{|\phi''(c)|}\lambda^{1/2}} + O(\lambda^{-3/2})\right)$$

as $\lambda \to \infty$.

Proof. Although the proof may be similar there is an important distinction in the solution. Suppose we wanted to try the same approach. We first assume, w.l.g., that c = 0. Then, we write

$$\begin{aligned}
\Phi(\lambda) &= \int_a^b g(t)\,\mathrm{e}^{-\lambda\phi(t)}\,\mathrm{d}t \\
&= \int_a^{-\delta} g(t)\,\mathrm{e}^{-\lambda\phi(t)}\,\mathrm{d}t + \int_{-\delta}^{\delta} g(t)\,\mathrm{e}^{-\lambda\phi(t)}\,\mathrm{d}t + \int_{\delta}^{b} g(t)\,\mathrm{e}^{-\lambda\phi(t)}\,\mathrm{d}t \\
&=: (1) + (2) + (3).
\end{aligned}$$

For (1) and (3) the result is as it was before

$$\begin{cases} |(1)| \leq \int_{\delta}^{a} |g(-t)|\,\mathrm{e}^{-\lambda\phi(-t)}\,\mathrm{d}t \leq \mathrm{e}^{\lambda\phi(-\delta)}\,||g||_{L^1[a,b]}, \\ |(3)| \leq \int_{\delta}^{b} |g(t)|\,\mathrm{e}^{-\lambda\phi(t)}\,\mathrm{d}t \leq \mathrm{e}^{\lambda\phi(\delta)}\,||g||_{L^1[a,b]}. \end{cases}$$

For (2) we try to define a function

$$H(t,s) := \phi(t) - \phi(0) - s^2.$$

However we can not apply Theorem 4.3 as the partial derivative $\partial_t H = \phi'$ vanishes at $t = 0$. A nice fact about this indetermination is that it does not come from the fact that there are no solutions to $H(t,s)=0$ but from the fact that there are two. To solve this we will use a technique called *blow up*. First, we introduce a variable $v := v(t)$ such that $vs = t$. Now, we redefine our function of interest to be

$$L(s,v) := \frac{\phi(sv)-\phi(0)}{s^2} - 1.$$

for $s \in (-\delta,\delta)-\{0\}$. We think this function is also singular but note that, using the Taylor Expansion of ϕ we write

$$\phi(t) = \frac{\phi''(0)}{2}t^2 + t^3 R(t)$$

for some error function R, which leave us with

$$L(s,v) = \frac{\phi''(0)}{2}v^2 + sv^3 R(sv) - 1,$$

and we can indeed apply the Implicit Function Theorem. In fact, we know $L \in C^2$ and as we want to solve for $L(s,v)=0$, we need that the v_0 we take to satisfy $L(0,v_0)=0$, ie,

$$v_0 = \sqrt{\frac{2}{\phi''(0)}}.$$

Now, we also know that

$$\partial_v L(0,v_0) = \phi''(0)v_0 \neq 0.$$

So we get a bijective and continuously differentiable function $v = v(s)$ such that

$$\begin{cases} L(s,v(s)) = 0 \\ v(0) = v_0 \end{cases}$$

But, $L(s,v(s))$ means exactly that

$$\phi(t)-\phi(0) = s^2$$

with $t = t(s) = sv(s)$. Getting back to our equation, we have that

$$\mathrm{e}^{\lambda\phi(0)}\int_{-\delta}^{\delta} g(t)\,\mathrm{e}^{-\lambda(\phi(t)-\phi(0))}\,\mathrm{d}t = \mathrm{e}^{\lambda\phi(0)}\int_{\alpha}^{\beta} g(sv(s))(v(s)+sv'(s))\,\mathrm{e}^{-\lambda s^2}\,\mathrm{d}s$$

where we calculate $\alpha = -\sqrt{\phi(-\delta)-\phi(0)}$ and $\beta = \sqrt{\phi(\delta)-\phi(0)}$. One can now again finish the proof with Proposition 4.1. □

Example 4.3. (NICA; ORTMANN, 2025) For $x \in \mathbb{R}$ fixed, i.e. $x = O(1)$, we have the following asymptotic as $n \to \infty$:

$$\mathrm{He}_n(x) = \sqrt{2}\left(\frac{n}{\mathrm{e}}\right)^{n/2} \mathrm{e}^{x^2/2} \cos\left(\frac{n\pi}{2} - x\sqrt{n}\right)(1+o(1)).$$

To see that, consider the integral representation of the Hermite Polynomial and re-write

$$\begin{aligned}
\mathrm{He}_n(x) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} (x+\mathrm{i}t)^n \,\mathrm{e}^{-t^2/2}\,\mathrm{d}t = \frac{2}{\sqrt{2\pi}} \Re\left\{\int_0^{\infty} (x+\mathrm{i}t)^n\,\mathrm{e}^{-t^2/2}\,\mathrm{d}t\right\} \\
&= \frac{2}{\sqrt{2\pi}} \Re\left\{\int_{\Re(z)=x, Im(z)>0} \mathrm{e}^{n\ln(z)+\frac{1}{2}(z-x)^2}\,\frac{\mathrm{d}z}{\mathrm{i}}\right\} \\
&= \frac{2}{\sqrt{2\pi}} \Re\left\{\int_{\Re(z)=0, Im(z)>0} \mathrm{e}^{n\ln(z)+\frac{1}{2}(z-x)^2}\,\frac{\mathrm{d}z}{\mathrm{i}}\right\}
\end{aligned}$$

where you start with a line integral in $\mathbb{C}$ and move the contour over to the positive imaginary axis - justified by the Cauchy's Theorem and the exponential decay of the integrand near infinity. Now, we focus on $n\ln(z) + \frac{1}{2}(z-x)^2$ which has maximum at $z^* = \mathrm{i}\sqrt{n} + o(\sqrt{n})$. The idea is to change variables by a factor of $\sqrt{z}$ so that the location of the maximum contribution stays $O(1)$ as n gets large. Specifically, we do

$$z = \mathrm{i}\sqrt{n}s, \qquad \frac{\mathrm{d}z}{\mathrm{i}} = \sqrt{n}\mathrm{d}s, \qquad \ln(z) = \ln(\sqrt{n}) + \ln(s) + \mathrm{i}\frac{\pi}{2}$$

which give us

$$\frac{1}{2}(z-x)^2 = -\frac{1}{2}ns^2 + \frac{1}{2}x^2 - \mathrm{i}\sqrt{n}sx$$

so that we remain with

$$\begin{aligned}
\mathrm{He}_n(x) &= \frac{2}{\sqrt{2\pi}} \Re\left\{\int_0^{\infty} \mathrm{e}^{n\ln(\sqrt{n})}\,\mathrm{e}^{n\ln(s)}\,\mathrm{e}^{n\mathrm{i}\frac{\pi}{2}}\,\mathrm{e}^{\frac{1}{2}x^2}\,\mathrm{e}^{-\mathrm{i}\sqrt{n}sx}\,\sqrt{n}\mathrm{d}s\right\} \\
&= \frac{2}{\sqrt{2\pi}} \sqrt{n}\,\mathrm{e}^{n\ln(\sqrt{n})}\,\mathrm{e}^{\frac{1}{2}x^2} \int_0^{\infty} \mathrm{e}^{n\left(\ln(s)-\frac{1}{2}s^2\right)} \cos\left(n\frac{\pi}{2} - \sqrt{n}sx\right)\mathrm{d}s,
\end{aligned}$$

where we can apply Laplace's approximation discussed before. Noting that $\phi(s) = \ln(s) - \frac{1}{2}s^2$ has $\phi'(s) = \frac{1}{s} - s$ and $\phi''(s) = -\frac{1}{s^2} - 1$, with global maximum at $s^* = 1$ with $\phi(1) = -\frac{1}{2}$ and $\phi''(1) = -2$, yielding

$$\begin{aligned}
\mathrm{He}_n(x) &= \frac{2}{\sqrt{2\pi}} \sqrt{n}\,\mathrm{e}^{n\ln(\sqrt{n})}\,\mathrm{e}^{\frac{1}{2}x^2} \left(\mathrm{e}^{n\phi(1)} \sqrt{\frac{2\pi}{n|\phi''(1)|}}\right) \cos\left(n\frac{\pi}{2} - \sqrt{n}x\right)(1+o(1)), \\
&= \sqrt{2}\,\mathrm{e}^{n\ln(\sqrt{n})}\,\mathrm{e}^{\frac{1}{2}(x^2-n)} \cos\left(n\frac{\pi}{2} - \sqrt{n}x\right)(1+o(1)), \\
&\sqrt{2}\left(\frac{n}{\mathrm{e}}\right)^{n/2} \mathrm{e}^{x^2/2} \cos\left(\frac{n\pi}{2} - x\sqrt{n}\right)(1+o(1)).
\end{aligned}$$

Remember that in **??** we mentioned the possibility of studying the distributions of eigenvalues from the asymptotic of the orthogonal polynomial. Not only that but in **??** we said that the Hermite polynomials are the orthogonal polynomials with respect to the Gaussian weight. Therefore, this asymptotic is directly related to the asymptotic of the distribution of eigenvalues of the GUE.

4.2.3 Steepest Descent

Let us now finally explore integrals over the complex plane with complex valued functions. We retake the integral

$$\phi(\lambda) := \int_\Gamma g(s)\,\mathrm{e}^{-\lambda f(s)}\,\mathrm{d}s, \tag{4.6}$$

where Γ is a contour in the complex plane and f, g are given complex-valued, analytic functions. Suppose that Γ is parametrized by $\gamma : [a,b] \to \mathbb{C}$ and let us decompose the functions we have into their real and imaginary parts such as

$$f(z) = u(z) + \mathrm{i}v(z), \quad g(z) = p(z) + \mathrm{i}q(z), \quad \gamma(t) = \alpha(t) + \mathrm{i}\beta(t)$$

with all functions in the decomposition being real valued. Of course we now have

$$\mathrm{e}^{-\lambda f(z)} = \mathrm{e}^{-\lambda u(z)}\,\mathrm{e}^{-\mathrm{i}\lambda v(z)},$$

and when λ grows large, we have the exponentially growing/decaying term $\mathrm{e}^{-\lambda u(z)}$ and a oscillating term coming from $\mathrm{e}^{\mathrm{i}\lambda v(z)}$ which is hard to control. The idea to get around this fact is to use paths in which this term is constant. If we have such a case, we write

$$\begin{aligned}\Phi(\lambda) = \mathrm{e}^{-\mathrm{i}\lambda v_0} &\int_a^b (p(\gamma(t))\alpha'(t) - q(\gamma(t))\beta'(t))\,\mathrm{e}^{-\lambda u(\gamma(t))}\,\mathrm{d}t \\ &+ \mathrm{i}\,\mathrm{e}^{-\mathrm{i}\lambda v_0} \int_a^b (p(\gamma(t))\beta'(t) - q(\gamma(t))\alpha'(t))\,\mathrm{e}^{-\lambda u(\gamma(t))}\,\mathrm{d}t \end{aligned} \tag{4.7}$$

where each integral can now be solved by Laplace's method.

Theorem 4.6. *(Steepest Descent) Consider Φ as in (4.6) with a smooth contour Γ. Assume that the exponent $f = u + \mathrm{i}v$ is an analytic function on a neighborhood of Γ that satisfies*

$$v(z) \equiv v_0, \quad z \in \Gamma$$

In addition, suppose that u has a unique minimum at an interior point of Γ, say on $z_0 \in \Gamma$, with $f''(z_0) \neq 0$, and also that g is analytic on a neighborhood of Γ and satisfies

$$\int_\Gamma |g(z)||\mathrm{d}z| < \infty.$$

Then Φ has a leading asymptotic as $\lambda \to +\infty$ given by

$$\Phi(\lambda) = \mathrm{e}^{-\lambda f(z_0)}\left(\frac{\sqrt{2\pi}g(z_0)\,\mathrm{e}^{\mathrm{i}\theta(z_0)}}{\lambda^{1/2}\sqrt{|f''(z_0)|}} + O(\lambda^{-3/2})\right), \quad \lambda \to +\infty,$$

where $\theta(z_0)$ is the angle of the oriented tangent of Γ at z_0.

Proof. Fix a parametrization $\gamma : [a,b] \to \mathbb{C}$ of Γ with $\gamma(c) = z_0, a < c < b$, and decompose Φ as before in (4.7). By the assumptions on the functions f, g we apply Laplace‘s method on both equations, and yields, as $\lambda \longrightarrow +\infty$,

$$\int_a^b (p(\gamma(t))\alpha'(t) - q(\gamma(t))\beta'(t))\, \mathrm{e}^{-\lambda u(\gamma(t))}\, \mathrm{d}t = \sqrt{2\pi}\, \mathrm{e}^{-\lambda u(z_0)} \left(\frac{p(z_0)\alpha'(c) - q(z_0)\beta'(c)}{\sqrt{|(u\circ\gamma)''(c)|}\lambda^{1/2}} + \epsilon \right)$$

and

$$\int_a^b (p(\gamma(t))\beta'(t) - q(\gamma(t))\alpha'(t))\, \mathrm{e}^{-\lambda u(\gamma(t))}\, \mathrm{d}t = \sqrt{2\pi}\, \mathrm{e}^{-\lambda u(z_0)} \left(\frac{p(z_0)\beta'(c) - q(z_0)\alpha'(c)}{\sqrt{|(u\circ\gamma)''(c)|}\lambda^{1/2}} + \epsilon \right)$$

with $\epsilon = O\left(\lambda^{-3/2}\right)$. Using that $v_0 + u(z_0) = f(z_0)$ and also that,

$$(p(z_0)\alpha'(c) - q(z_0)\beta'(c)) + \mathrm{i}(p(z_0)\beta'(c) - q(z_0)\alpha'(c)) = g(z_0)\gamma'(c),$$

and, combining the estimates for both parts we get

$$\Phi(\lambda) = \sqrt{2\pi}\, \mathrm{e}^{-\lambda f(z_0)} \left(\frac{g(z_0)\gamma'(c)}{\sqrt{|(u\circ\gamma)''(c)|}\lambda^{1/2}} + O\left(\lambda^{3/2}\right) \right)$$

and it suffices now to get an expression for the second derivative in the denominator. Knowing the imaginary part is constant we write

$$\frac{\mathrm{d}}{\mathrm{d}t} u(\gamma(t)) = \frac{\mathrm{d}}{\mathrm{d}t} f(\gamma(t)) = \gamma'(t) f'(\gamma(t))$$

and

$$\frac{\mathrm{d}^2}{\mathrm{d}t} u(\gamma(t)) = \frac{\mathrm{d}^2}{\mathrm{d}t} f(\gamma(t)) = (\gamma'(t))^2 f''(\gamma(t)) + \gamma''(t) f'(\gamma(t)).$$

Hence,

$$\frac{\gamma'(c)}{\sqrt{|(u\circ\gamma)''(c)|}} = \frac{1}{\sqrt{|f''(z_0)|}} \frac{\gamma'(c)}{|\gamma'(c)|}.$$

□

We now need do discuss why this assumption of constant value of I$m(f(z))$ is not an assumption as strong as one might assume at first. Suppose we want to integrate in a arbitrary contour Γ of the plane. Of course, our assumption that the functions were analytic permit us to deform the contour keeping the value of the integral. We are going to use this to find a contour $\tilde{\Gamma}$ such that sections of the contour are such that I$m(f(z))$ is constant. We already know that being analytic means satisfying the *Cauchy-Riemann* equations, $u_x(z) = v_y(z), \text{and} u_y(z) = -v_x(z)$. An immediate result from these equations is that the gradients of u and v must be normal. Indeed, if $\langle\cdot,\cdot\rangle$ is the inner product on $\mathbb{R}^2$,

$$\langle \nabla u(z), \nabla v(z)\rangle = u_x(z)v_x(z) + u_y(z)v_y(z) = u_x(z)v_x(z) - v_x(z)u_x(z) = 0.$$

As we are following the curves where I$m(f(z))$ is constant, we are also following the path of Steepest Descent of the real part of the integral. As we also are looking to use Laplace's method on the integrals, we may want to have our paths going through the critical points of the function. This fully motivates the method described.

Example 4.4. There are three asymptotics regimes for $\mathrm{He}_n(x)$ when $x \to \infty$ depending on the value of $\lim_{N\to\infty} x/2\sqrt{N}$ for $N = n+1$, each of which has qualitatively different behavior

- Small x values $-2\sqrt{N} < x < 2\sqrt{N}$: in this regime the Hermite Polynomials will be oscillating, and a have a similar qualitative behavior as in Example 4.3, with modification to phase and amplitude.
- Large x values $|x| > 2\sqrt{N}$: In this regime the polynomial does not have more oscillations and is monotone increasing. As x grows larger, this approximation behaves as $x^N \mathrm{e}^{-N^2/2x^2}$.
- Edge x values $x = \pm 2\sqrt{N} + O(N^{-\frac{1}{6}})$: in this regime the polynomials will be approximated by the Airy Function. This interpolates between the oscillatory small x regime and the monotone large x regimes.

We write the integral representation given by:

$$\begin{aligned}\mathrm{He}_n(x) &:= \frac{n!}{2\pi\mathrm{i}}\int_\Gamma z^{-n-1}\,\mathrm{e}^{xz-\frac{z^2}{2}}\,\mathrm{d}z\\ &= \frac{n!}{2\pi\mathrm{i}}\int_\Gamma \mathrm{e}^{xz-\frac{z^2}{2}-(n+1)\ln(z)}\,\mathrm{d}z\end{aligned}$$

where Γ is a contour that goes around the origin coming and going from minus infinity such as $\Gamma = \{\lambda - \mathrm{i}\delta : \lambda < 0\} \cup \{\delta\,\mathrm{e}^{\mathrm{i}\theta} : -\pi/2 < \theta < \pi/2\} \cup \{\lambda + \mathrm{i}\delta : \lambda < 0\}$. This is equivalent to using $\Gamma = \{\mathrm{e}^{\mathrm{i}\theta} : 0 \le \theta < 2\pi\}$. To determine He_n, set $N = n+1$ and consider the scaling $x \mapsto N^{1/2}\alpha$ and $z = N^{1/2}\beta$

$$\begin{aligned}\mathrm{He}_n(N^{\frac{1}{2}}\alpha) &= \frac{(N-1)!}{2\pi\mathrm{i}}\int_\Gamma \mathrm{e}^{(N^{1/2}\alpha)(N^{1/2}\beta)-\frac{1}{2}(N^{1/2}\beta)^2-N\ln(N^{1/2}\beta)}\,\mathrm{d}\beta\\ &= \frac{(N-1)!N^{\frac{1}{2}(1-N)}}{2\pi\mathrm{i}}\int_\Gamma \mathrm{e}^{-N(-\alpha\beta+\frac{1}{2}\beta^2+\ln(\beta))}\,\mathrm{d}\beta,\end{aligned}$$

so that we have

$$\mathrm{He}_n(\sqrt{N}\alpha) = C_n^0\int_\Gamma \mathrm{e}^{-N\phi_\alpha(\beta)}\,\mathrm{d}\beta, \quad \phi_\alpha(\beta) = \frac{1}{2}\beta^2 - \alpha\beta + \ln(\beta).$$

As we want to calculate the asymptotic by steepest descent, we need to determine the critical points of such a function and then the steepest descent paths passing through this points. We can easily determine the critical points as $\beta_c^\pm = \frac{\alpha\pm\sqrt{\alpha^2-4}}{2}$ - solutions to the quadratic formula given by solving the critical point of the function (the zeros of its derivatives). Now, the behavior of the critical points will depend on the interval of the real parameter α,

$$\beta_c^\pm = \frac{\alpha\pm\sqrt{\alpha^2-4}}{2} = \begin{cases}\mathrm{e}^{\pm\gamma}, & \alpha = 2\cosh(\gamma) > 2, \quad \gamma > 0;\\ \mathrm{e}^{\pm\mathrm{i}\gamma}, & \alpha = 2\cos(\gamma) \in [0,2), \quad \gamma \in (0,\pi/2];\\ 1, & \alpha = 2,\end{cases}$$

each of the cases refers to one of the regimes we described before.

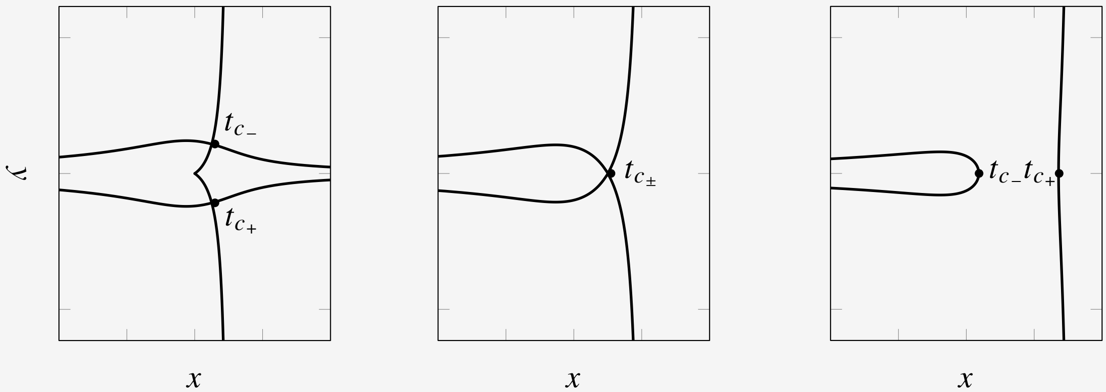

Figure 2 – The lines represent the solution to $\mathrm{I}m[f(z)-f(t_c)]=0$ for the three possible cases of the Hermite Polynomial. the cases are from left to right $\alpha \in [0,2)$, $\alpha = 2$ and $\alpha > 2$. Only for the first case the critical points are complex. These are the curves we use to perform the asymptotic analysis on the integral. Note the case $\alpha = 2$; the coalescence of critical point gives origin to the Airy function.

CHAPTER

5

FREE ENERGY

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